Color and Perception

Notes on the work of V. Zheleznyakov

Based on "Color and Contrast. Technology and Creative Choice"

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Introduction

- **HUE** [violet: red] - angular displacement
- **SATURATION** [achromatic: pure tone] - displacement from the center of the color wheel to the edge
- **LUMINANCE/BRIGHTNESS** [0: 1] - range moves

Color Spaces

- RGB (Red, Green, Blue) – monitor, video camera
- CMY (Cyan, Magenta, Yellow) – printing model
- HSL (Hue, Luminance, Saturation) – close to human perception of color

Here you can visually see the change in color when changing parameters in several color models: https://www.hslpicker.com/

HSL model

1. Dip and whitening of tone when changing lightness (see Figure 1).
2. Change in color saturation when moving along the radius from the center (achromatic shade of gray) to the edge (pure tone) (see Figure 2).

Guideline

> "The point, of course, is not which scheme is more accurate - let's leave that to psychologists, the point is that the visible and the known are intertwined in our consciousness in the most bizarre way. This is the leitmotif of the book."
>
> — *"Color and Contrast. Technology and Creative Choice"* V. Zheleznyakov

1 Object and its Image

Despite differences in technique, cinema and painting have a common ground. These are the laws of psychophysiological perception of color; first the color of the object, and then its image. Only the term "color" should be understood more broadly: not as the color of individual objects (sky, earth, greenery, face, costume), but as a single color environment with multi-colored lights, shadows, reflections, and highlights, united by the unity of place and lighting. That is, color not in the colorimetric, but in the picturesque sense.

It is known that an image cannot be absolutely similar to an object, but speaking of the dissimilarity between an object and its image, three main reasons should be mentioned:

1. The reproducing system, which introduces large changes in the image compared to the object.
2. Inadequacy of receptive sensation and perceptual perception.
3. The difference between the image of the object and the viewer's idea of this object.

The duplication theory distinguishes three degrees of similarity in color between an object and its image:

- **Physically accurate similarity**
- **Physiologically accurate similarity**
- **Psychologically accurate similarity**[1]

The image is what we feel.

The image is the result of psychological, not just physiological, perception.

The main problems associated with color in the frame do not depend on the technical properties of the reproducing system, but on the relationship between the sensory response of our body and the idea of what we see, i.e. they are the result of the dissimilarity between what we see and how we perceive it. Moreover, this situation applies equally to the perception of the object itself and to the perception of its image. Different types of fine art have their own specific techniques (forms) that make the viewer perceive the image in the same way as the object.

The image of an object on the movie screen is an object for the viewer.

2 Sensation and Perception

Camera → reproduction. Eye → visual image.

Visual perception is not a mechanical registration of sensory elements, it turns out to be a truly creative ability of instantaneous grasping of reality, an imaginative and inventive ability. Perception is at the same time thinking, any reasoning is at the same time intuition, and any observation is also creativity.

Perception is not a mechanical registration of elements, but the grasping and comprehension of significant structural properties of an object. Each act of perception is a visual judgment.

2.1 Gestalt psychology

(for cinematography, photography, for the visual part).

We are considering Gestalt psychology for cinema, so I will be very brief and describe only the main idea, as well as the principles that are important for the perception of the image.

To begin with, let's talk about the position in which human perception is divided into several stages: sensory, perception of content, perception taking into account the environment,

the culture of society, if you like, in which a person is. Next will be about the first two stages, however, the director should take the last one very seriously. Also, one should not forget about the psyche, which is individual for each person, this is his individual "prism of perception", which significantly affects emotional perception. Emotional perception is a synthesis of all stages of perception, which were discussed above, and the individual "prism of perception" imposed by the psyche. Art is able to convey emotion, this is the main goal and value, but this emotional part literally does not exist before the viewer begins to perceive the work.

2.1.1 1. Structural principle

The structural principle is important for the first stage of perception - sensory perception. Our perception is arranged in such a way that it strives for harmonization, ordering (we build space, objects, in a word - the known from lines, spots, contours of spots). Only after the ordering process, when we understood what is in front of us, can we move on to the second stage of perception, the stage of content perception.

Here I think it will not be superfluous to introduce the concept of Arnheim's entropy model of perception. For now, I will talk about this model in the context of the first stage. The structure of the image, which the image has, has a certain level of complexity for ordering by our sensory perception. This level of complexity is entropy, or a measure of chaos. Entropy is maximal at the beginning of the perception stage and decreases as harmonization proceeds. If the measure of entropy is too low, then in sensory perception, the structure of the image (form) does not cause a nervous rise, in other words, interest. Too high entropy, as well as too low, can prevent the transition from the first stage of perception to the second. We should try to introduce as much entropy into the structure of the image so that it does not become too complex or easy to harmonize. The viewer, as it were, completes the image.

The same entropy model also works for the perception of content, which can be conditionally divided into "new" and "known". We should try to balance this wisely. Despite the fact that here it is complicated by the fact that what is "new" for one viewer may well be "known" for another.

2.1.2 2. The principle of constancy

The essence of the phenomenon of constancy lies in the ability of the entire visual system to take into account and compensate for changes in the retinal image that occur with changes in distance or color of illumination. In other words, constancy is the adaptation of the perceptually perceived to the known, or the substitution of the visible with the known. There is constancy in color, form, lightness, etc.

What color is the grass? What color is the brickwork? Green? Red? Here is an example of constancy. Yes, here we can say that we see what we want to see.

2.1.3 3. The principle of gradient balance (perceptual gradient)

A gradient is a measure of change. The magnitude of the change in anything: lightness, size, angle, shooting point, composition, tone, saturation, speed of movement of the camera and objects in the frame, and so on. Gradients are important because they are the ones that create contrasts, analogies, rhymes, rhythms, help the viewer to navigate what he sees (semantic markers, more on them later).

The provisions of Gestalt psychology that describe the perception of "simple forms", such as, for example, the relationship between figure and background:

1. The figure must be clearly separated from the background.
2. A smaller formation tends to be perceived as a figure, a larger one as a background, especially if the composition is not very clear in the frame and the objects are unfamiliar.

3. As a rule, similar visual elements located close to each other are perceived as figures, not as a background.

4. Symmetrical and filled formations usually tend to be perceived as a figure as well.

According to Gestalt psychology, our perception strives for structuring (the search for simplification, repeatability, purity, clarity and orderliness). The consequence of this is that the simplest formations in the image have a greater chance of being perceived as a figure. The sometimes-arising uncertainty of the "figure-background" relationship when building a composition has the right to exist, but it loses in expressiveness.

It is also worth mentioning the impossibility of simultaneous perception of the figure and the background.

A diagram that reflects the dependence of the viewer's "interest" on the balance of the new and the known in the work, this is also applicable to the content, and not just to the image (see Figure 3).

3 Features of Perceptual Color Perception

It is impossible to comprehend the details of texture and volumetric form and at the same time feel all the subtlest nuances of color in lights, shadows, highlights and reflections.

Perception proceeds as if through two different channels, and to switch attention from one to the other requires some internal effort and even a certain skill (professional painters have this skill).

This happens due to the fact that when we perceive an object as a whole, we get some general idea of the color of its certain brightness zones (modulations: lights, shadows, highlights, reflections), which is enough for the constancy mechanism to complete our idea of the true color of the object. But if we proceed to a more careful analysis of the color structure (we try to see the change in color when moving from one modulation to another), focusing only on color differences, then our gaze ceases to distinguish the features of form and texture, focusing only on color differences. During this process, our eye changes its lightness adaptation in such a way that the colors being compared at the moment of attention are of such lightness at which color differences are most noticeable for a given tone. It is also worth mentioning that to compare one color with another, it is necessary to bring the colors to a single lightness value, otherwise the eye will adapt to a lighter area, darkening the darker one, which will lead to a loss of color in this area.

3.1 Color intervals

According to N.N. Volkov, we distinguish two types of color interval systems:

- **Large intervals** are the boundaries between colored objects within the selected composition (often used for planar painting with local color).

- **Small intervals** are shades of close colors within one object.

The visual perceptual gradient in the perception of lightness differences of the same object can change depending on the zone of our attention, i.e. depending on what we currently consider to be narratively important. When considering a separate dark part of the image, for our perception, the size of the intervals in it will be higher than in the light part, due to the adaptation of the eye.

3.1.1 Color series

Color intervals form color series, which are almost never equal-stepped, at least due to the change in color according to different parameters. Color series can, for example, convey the direction and acceleration of movement towards a color accent and, conversely, a slowdown when moving away from it. In color series, the contrasting influence of neighboring spots is widely used, but the effect of color influences in richly and rhythmically constructed series goes far beyond the phenomenon of color contrast studied in physiology. The movement of color is able to generate in the alternation of almost neutral warm and cold spots the impression of full color ("St. Sebastian" by Titian).

Several color series, intertwining with each other in a special way, form the color structure of the work.

> "The mutual influence of colors in a coloristically constructed painting far exceeds, both in the magnitude of changes and in complexity, everything that has been studied so far by color scientists and psychologists under the headings of simultaneous and successive contrast."
>
> — N.N. Volkov *"Color in Painting"*

3.1.2 Replacing contrasts

"Boy with a Dog", Pablo Picasso - the color contrast replaces the almost absent lightness contrast (see Figure 4).

3.1.3 Strengthening lightness contrast with color contrast

"Vuzovki", Istomin - the lightness contrast between the window and the walls is enhanced by the color contrast (greenish walls, a slight purple tint of the window) (see Figure 5).

When analyzing the problem of the similarity of the color of an object and its image in a work of fine art, one should not forget about the conscious, creative change of it for greater expressiveness. In this case, is it legitimate to raise the question of the admissibility of any restrictions for the artist? Valentin Nikolaevich Zheleznyakov believes that here everyone determines the limit for himself and gives the judgment that any work should be evaluated according to the laws that it itself proclaims, and the only criterion can only be the emotional response of the viewer to the author's message. Therefore, the material offered by the author is more likely to give the reader a reason for independent reflection and search than to impose any ready-made solutions.

Literature

- Zheleznyakov V.N. "Color and contrast. Technology and creative choice." M.: VGIK, 2001
- Arnheim R. "Art and Visual Perception"

Notes

1. In practice, it is with this that we always deal. It should be recalled that subtractive synthesis in the formation of colors on a three-layer positive film in cinema, on photographic paper in photography, and even in oil painting cannot, for various reasons, provide physiologically accurate similarity, but psychologically accurate similarity suits us quite well. The same applies to additive conversion in television.

A Illustrations

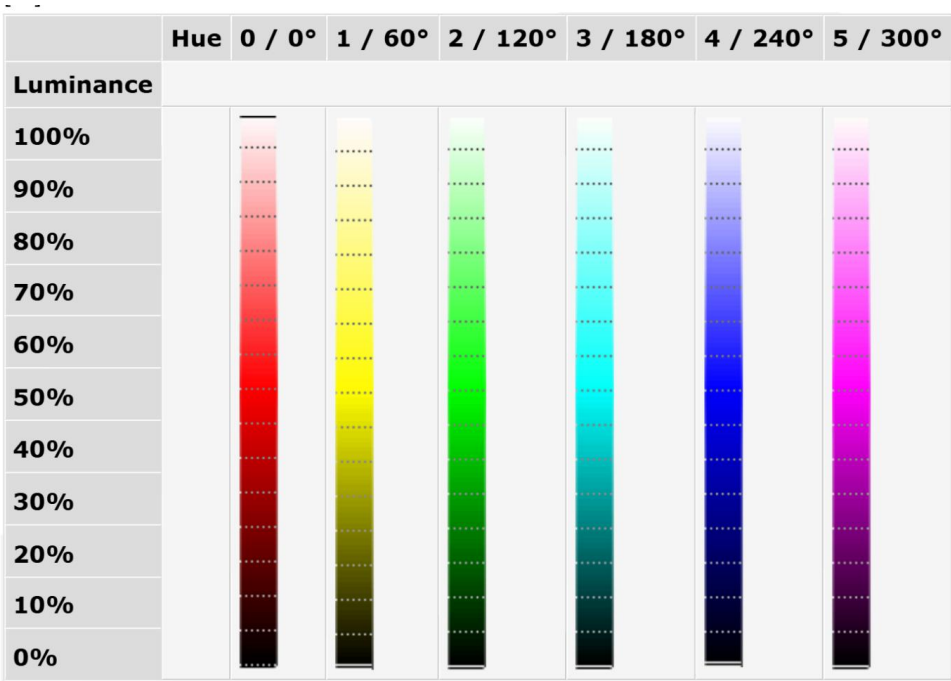

Figure 1: Dip and whitening of tone when changing lightness.

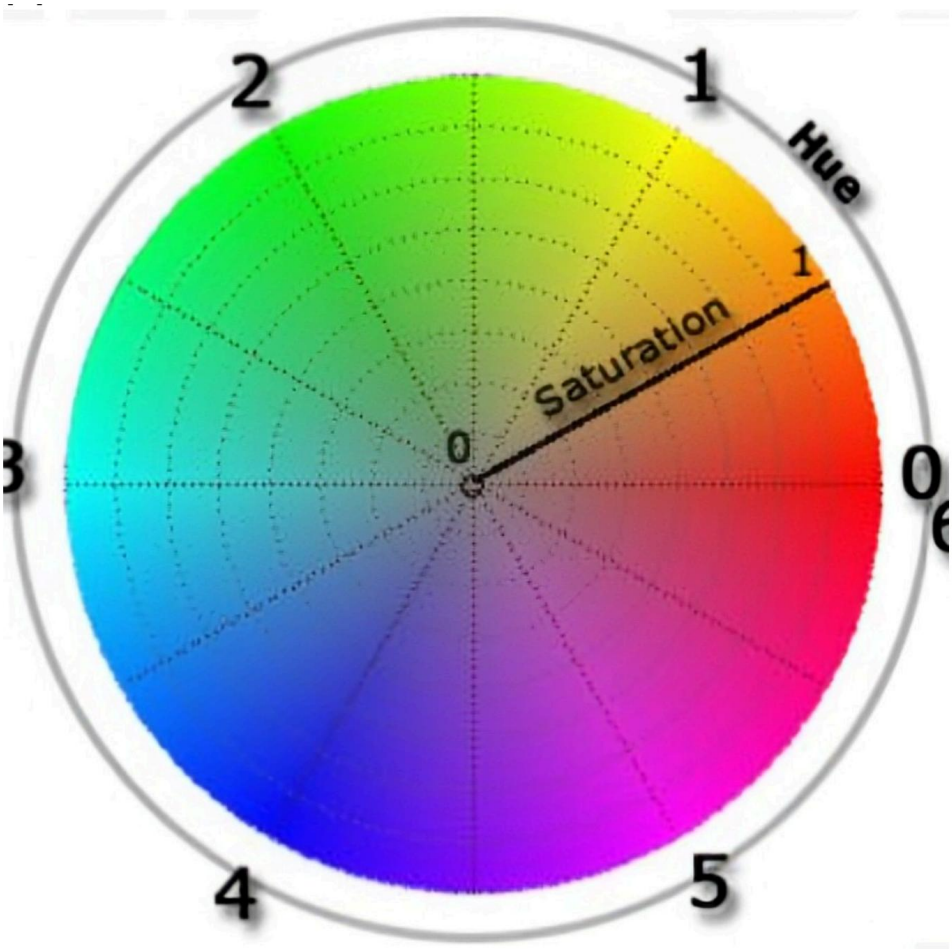

Figure 2: Change in color saturation when moving along the radius from the center to the edge.

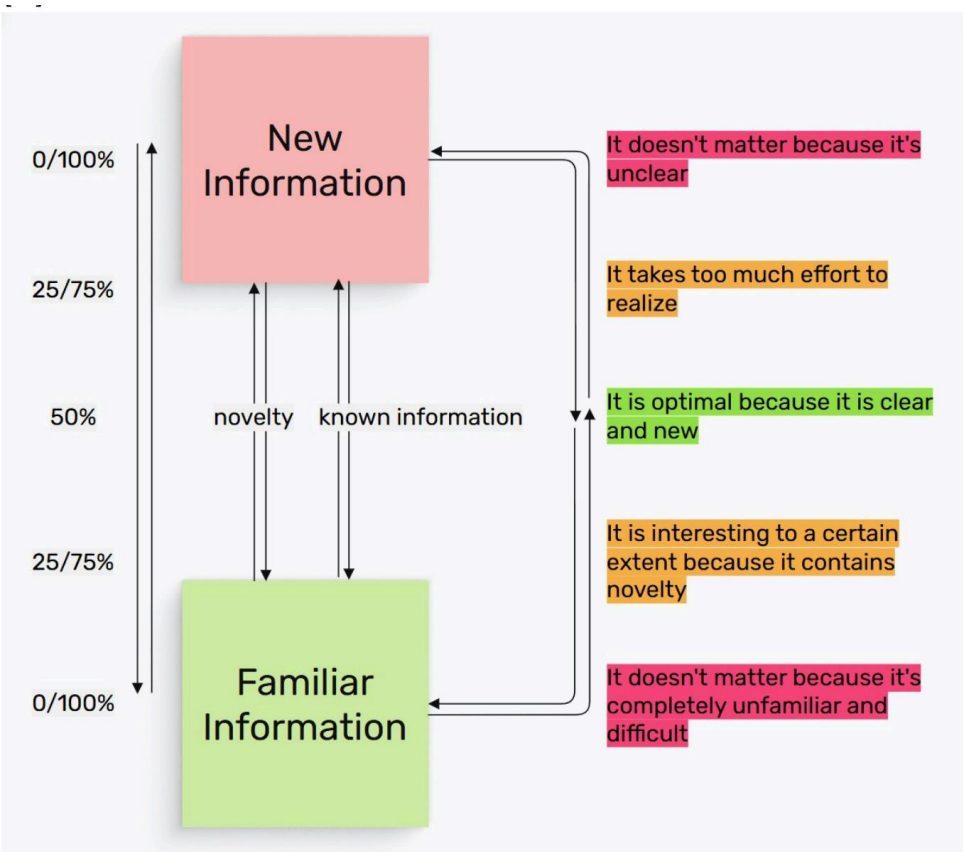

Figure 3: A diagram that reflects the dependence of the viewer's "interest" on the balance of the new and the known in the work.

Figure 4: "Boy with a Dog", Pablo Picasso - color contrast replaces lightness contrast.

Figure 5: "Vuzovki", Istomin - lightness contrast is enhanced by color contrast.